

LOOPS IN JAVA

WHILE LOOP

1. Print Numbers from 1 to N

Write a program to print numbers from 1 to N using a `while` loop.

2. Sum of Natural Numbers

Write a program to calculate the sum of natural numbers from 1 to N.

3. Sum of 5 User-Entered Numbers

Write a program to take 5 numbers from the user and print their sum.

4. Compute Series: $s = x + x^2 + x^3 + \dots + x^n$

Write a program to find the value of $s = x + x^2 + x^3 + \dots + x^n$, where x and n are entered by the user.

5. Compute Series: $s = x + x^2/2 + x^3/3 + \dots + x^n/n$

Write a program to calculate the sum of the series: $s = x + x^2/2 + x^3/3 + \dots + x^n/n$.

6. Multiplication Table Generator

Write a program to print the multiplication table of a given number.

7. Product of 5 User-Entered Numbers

Write a program to take 5 numbers as input and print their product.

8. Factorial of a Number

Write a program to calculate the factorial of a number entered by the user.

9. Sum of Digits of a Number

Write a program to calculate the sum of digits of a number entered by the user.

10. Compute Series: $s = x + x^3/3 + x^5/5 + x^7/7 + \dots + x^n/n$

Write a program to evaluate the series $s = x + x^3/3 + x^5/5 + x^7/7 + \dots + x^n/n$.

11. Compute Alternating Series: $s = 1 + x - x^2/2 + x^3/3 - x^4/4 + \dots + x^n/n$

Write a program to find the value of the alternating series: $s = 1 + x - x^2/2 + x^3/3 - x^4/4 + \dots + x^n/n$.

12. Reverse a Number

Write a program to reverse the digits of an entered number.

13. Check for Armstrong Number

Write a program to check whether a number is an Armstrong number. (Example: $153 = 1^3 + 5^3 + 3^3 = 153$)

14. Check for Strong Number

Write a program to check whether a number is a Strong number.

(A Strong number is a number for which the sum of the factorial of its digits equals the number itself. Example: $145 = 1! + 4! + 5! = 145$)

15. Check for Palindrome Number

Write a program to check whether a number is a palindrome.

(A number is a palindrome if it reads the same backward as forward, e.g., 121)

16. Check for Prime Number

Write a program to check whether a number is prime or not.

17. Check for Perfect Number

Write a program to check whether a number is a Perfect number.

(A Perfect number is equal to the sum of its proper divisors, e.g., $28 = 1 + 2 + 4 + 7 + 14$)

DO WHILE LOOP

1. Print Numbers from 1 to N using do-while loop

Write a program to print numbers from 1 to N using a do-while loop.

2. Print All Even Numbers from 2 to N

Use a do-while loop to print all even numbers from 2 up to N.

3. Find the Largest Digit in a Number

Use a do-while loop to find the largest digit present in a given integer.

4. Check if a Number is a Perfect Square

Using a do-while loop, check whether a given number is a perfect square (without using built-in sqrt functions).

5. Reverse a Number

Write a program to reverse the digits of an entered number.

6. Check for Armstrong Number

Write a program to check whether a number is an Armstrong number.
(Example: $153 = 1^3 + 5^3 + 3^3 = 153$)

CodeWithNishchal